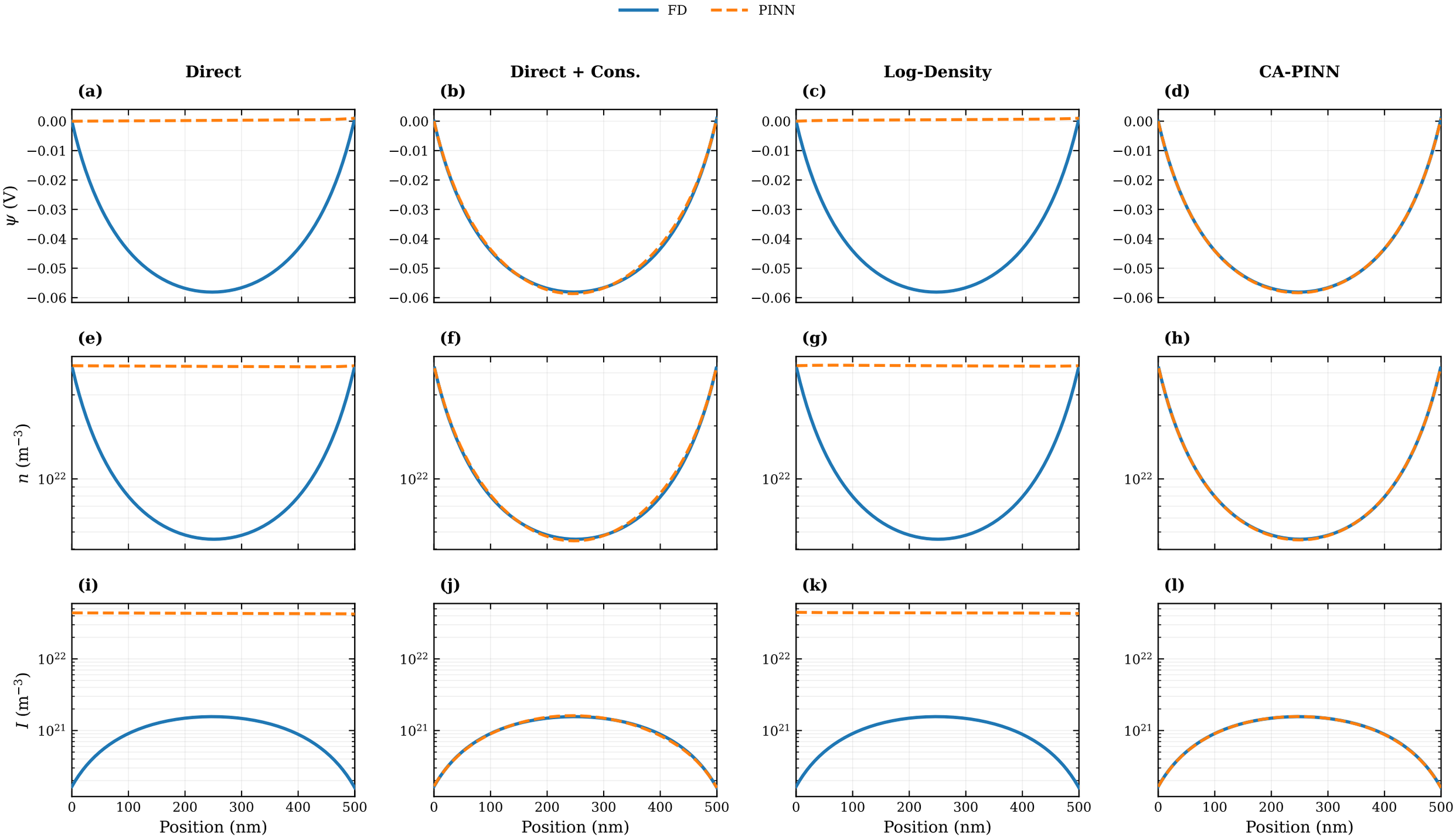
FD
PINN
Direct
Direct + Cons.
Log-Density
CA-PINN
(a)
(b)
(c)
(d)
(e)
(f)
(g)
(h)
(i)
(j)
(k)
(l)
ψ (V)
n (m⁻³)
I (m⁻³)
0.00
−0.01
−0.02
−0.03
−0.04
−0.05
−0.06
10²²
10²¹
0
100
200
300
400
500
Position (nm)